

HAT-P-32b

R-0185

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-32_241018 · created 2026-07-16T19:50:52+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460601.727 +/- 0.0031 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.027
Transit depth (Rp/Rs)^2	3.43 +/- 0.98 [%]
Orbital Inclination (inc)	89.66 +/- 1.53
Airmass coefficient 1 (a1)	1.011 +/- 0.03
Airmass coefficient 2 (a2)	-0.005 +/- 0.059
Scatter in the residuals of the lightcurve fit is	3.01 %
Best Comparison Star	#1 - [571, 51]
Optimal Method	PSF photometry
Transit Duration (day)	0.1027 +/- 0.0035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.027 vs 0.1489 (deviation 1.33σ)
Duration fit vs published	0.1027 d vs 0.1304 d (deviation 7.87σ)
Mid-time O–C	9.5 min
Depth SNR	6.8
Residual scatter	3.01 %

Light curve

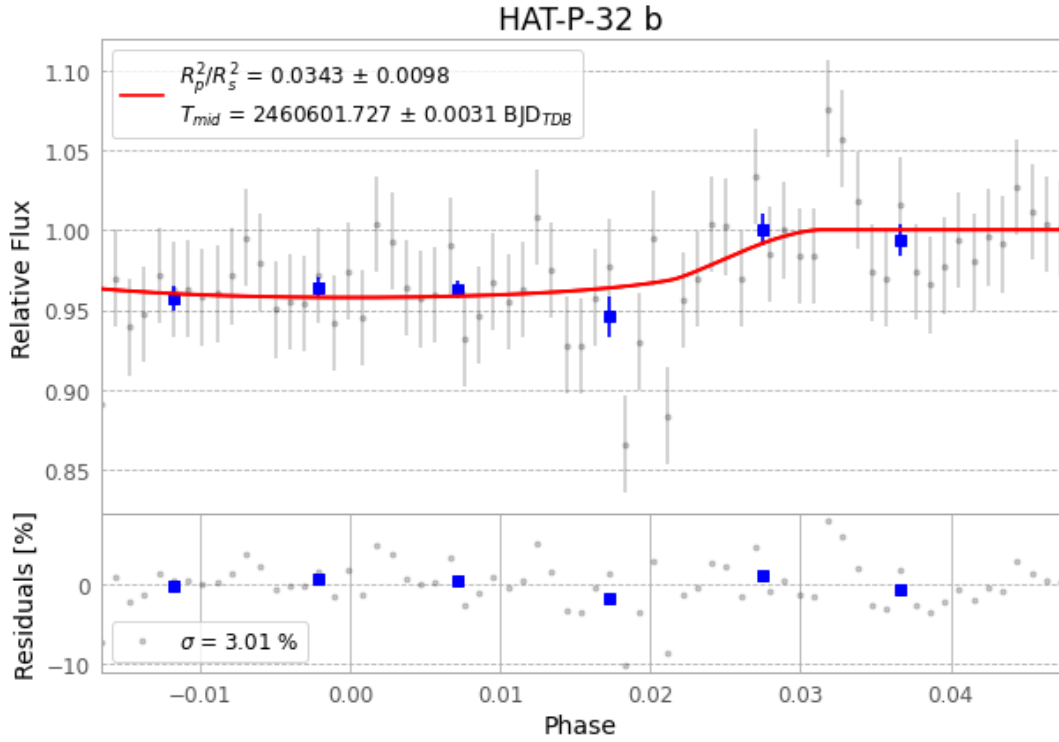


Figure 1 — FinalLightCurve_HAT-P-32 b_2024-10-17.png (SHA-256 92e17388825cbba0...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [367, 208], 1 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 11851fa08b3e3f59...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ee7e3ca

Data provenance

67 FITS frames (44 MB), HATP-32241018042715.FITS → HATP-32241018074515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-241018.log (c078b9abc984...), FinalLightCurve_HAT-P-32 b_2024-10-17.png (92e17388825c...), FinalLightCurve_HAT-P-32 b_2024-10-17.csv (1c99ad553f68...)

Compute resources

Wall time 125.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 19:50Z.